Always Valid Inference

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1 (FREQUENTIST) STATIC HYPOTHESIS TESTING PRECURSORS

Consider A real-valued random vector $X \sim P_\theta$ with support on $\mathbb{R}^n$ for some unknown $\theta \in \Theta$. We partition $\Theta = \Theta_N \sqcup \Theta_A$ and we wish to test the hypothesis that $\theta \in \Theta_N$. We call $\theta \in \Theta_N$ the *null hypothesis* and $\theta \in \Theta_A$ the *alternative hypothesis*.

Definition 1 (Hypothesis Test)**.** A *hypothesis test* is a function $\delta : \mathbb{R}^n \to \{N, A\}$ that for a given value of its input decides whether to pick the alternative in favor of the null or the null in favor of the alternative.[a]

[a] Two comments here: (1) typically when we pick the alternative hypothesis in favor of the null hypothesis, we say that we *reject the null hypothesis*. When we pick the *null hypothesis* in favor of the alternative hypothesis, we say we *fail to reject the null*. (2) In reality, what's defined here is a *non-randomized hypothesis test*. For any or all observations, a test is allowed to randomize between assigning N and A so as the maximize error trade off as discussed below. The randomization aspect and its motives will be ignored for brevity.

There are two types of errors that we can make when conducting a hypothesis test:

(1) **Type I error**: we favor the alternative hypothesis when the null holds. Mathematically, $\theta \in \Theta_N$ but for $X \sim P_\theta$, $\delta(X) = A$.

(2) **Type II error**: We favor the null hypothesis when the alternative holds. Mathematically, $\theta' \in \Theta_A$ but for $X \sim P_\theta$, $\delta(X) = N$.

We want to conduct the hypothesis test to keep these probabilities as small as possible. It's difficult to simultaneously control both error rates so typically we proceed as follows to choose the test δ:

(1) We choose $\alpha \in (0, 1)$, which we call the *level* of the test. That is the maximum Type I error we're willing to tolerate.

(2) We solve the problem

$$\begin{aligned} \delta_\alpha := \arg\min_{\delta} \sup_{\theta \in \Theta_A} & P_\theta(\delta(X) = N) \\ \text{s.t. } & P_\theta(\delta(X) = A) \leq \alpha \; \forall \theta \in \Theta_N \end{aligned}$$

This expression is dense. To unpack it, for a given level α, we choose the test δ to minimize the biggest likelihood that we incorrectly favor the alternative hypothesis to the null hypothesis subject to keeping the likelihood that we incorrectly favor the null hypothesis to the alternative hypothesis less than or equal to α. We call δ_α the *uniformly most powerful test* at level α.

δ_α can be characterized by two regions N_α and A_α where $N_\alpha \sqcup A_\alpha = \mathbb{R}$. The characterization is in the sense that

$$\begin{aligned} \delta_\alpha(X) = N &\iff X \in N_\alpha \\ \delta_\alpha(X) = A &\iff X \in A_\alpha \end{aligned}$$

so that N_α gives the region of X values where we favor the null hypothesis to the alternative hypothesis and A_α gives the complimentary region.

Often times, these tests take the form that $A_\alpha \subset A_{\alpha'}$ for $\alpha < \alpha'$. That is, whenever we reject the test at a lower level, we also reject the test at a higher level. In these cases, it's useful to report a p-value.

Definition 2 (p-value)**.** The p-value is a function $p : \mathbb{R} \to [0, 1]$ where

$$p(X) := \inf\{\alpha \in [0, 1] : X \in A_\alpha\}$$

That is, the p-value is the *smallest* level at which we would favor the alternative hypothesis in favor of the null hypothesis.[a]

[a] The reason that we define the p-value using the $\inf$ notation is so that we've defined a sort of generalized inverse. When each hypothesis has a single distribution with a density, the definition of the p-value becomes more clear and involves an inverse; see Section 1.1 for an example.

Example 3 (p-value Example)**.** As an example, suppose that $p(X) < \gamma$ for some $\gamma \in [0, 1]$. Then, we know that $\inf\{\alpha \in [0, 1] : X \in A_\alpha\} < \gamma$. That means that $A_{p(X)} \subset A_\gamma$ and since $X \in A_{p(X)}$ by definition, we have that $X \in A_\gamma$. In words, we favor the alternative to the null at level γ.

1.1 An Example and Intuitive Commentary

Consider the example in Figure 1. We observe X on the real-line and wish to decide whether it came from hypothesis N or hypothesis A, which are two different Gaussian distributions. Very intuitively speaking, if we observe $X = x_1$, then we'll generally want to favor hypothesis N to hypothesis A but if we observe $X = x_2$, then we'll generally want to favor hypothesis A to hypothesis N. There's a gray area in the middle where it's less clear which hypothesis to pick. Next, we give some example tests

(a) Consider $\delta = N$. This test ignores the data and *always* favors the null hypothesis to the alternative hypothesis. In this case, the probability of a Type I error is 0 and the probability of a Type II error is 1.

(b) Consider $\delta = A$. This test ignores the data and *always* favors the alternative hypothesis to the null hypothesis. In this case, the probability of a Type I error is 1 and the probability of a Type II error is 0.

(c) The most powerful tests in this case take the form $\delta(X) = \begin{cases} N, & \text{if } X \leq \kappa, \\ A, & \text{if } X > \kappa \end{cases}$ for $\kappa \in \mathbb{R}$. In fact, for any given level to the test α, we can define the function $\kappa : [0, 1] \to \mathbb{R}$ that takes in the level of the test and specifies the region $A_\alpha := \{x \in \mathbb{R} : x \geq \kappa(\alpha)\}$.

For $\alpha < \alpha'$, we convince ourselves that $\kappa(\alpha) > \kappa(\alpha') \implies A_\alpha \subset A_{\alpha'}$ so that we reside in a regime where p-values are useful. In fact, since the two hypotheses are Gaussian distributions which have densities, we have that κ is a strictly increasing function of α so that it has an inverse. We have that for any $X \in \mathbb{R}$, $p(X) = \inf\{\alpha \in [0, 1] : X \in A_\alpha\} = \kappa^{-1}(X)$.

2 Always Valid Inference

2.1 Motivation

In the industry, practitioners conduct hypothesis tests, which are colloquially called A/B tests, for product experimentation. For instance, an online retailer may wish to test between which website layout leads to more purchases.

Analysts continuously monitor results from the A/B tests over time with the sample size increasing as shown in Figure 3 taken from Johari et al. (2022). On the x-axis is time and on the y-axis we plot $1 - p_n$ where $p_n := p_n(X_n)$, X_n is the sample at time n, and p_n is the p-value as defined in Definition 2 associated with the static hypothesis testing problem at time n. With a given level α, where $1 - \alpha = 0.95$ is plotted in red, we observe that $1 - p_n > 1 - \alpha$ for two periods over

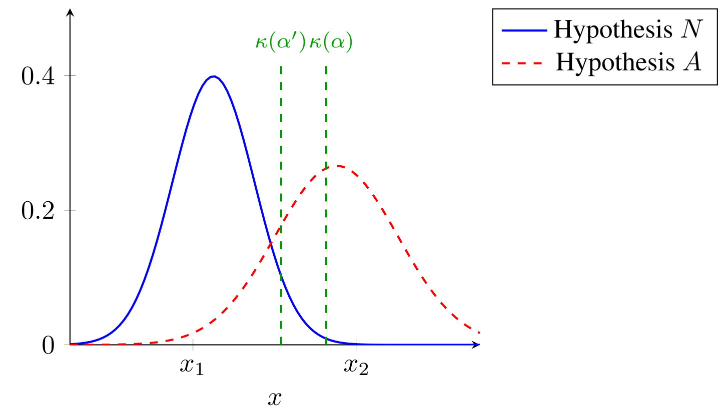

Figure 1. Hypothesis Testing Example

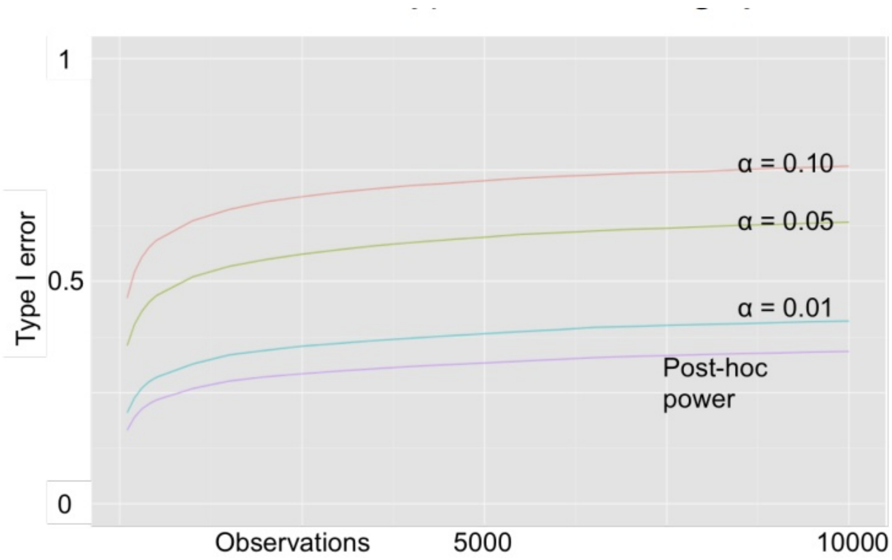

Figure 2. Continuous Monitoring of Static p-values Inflated Type I Error

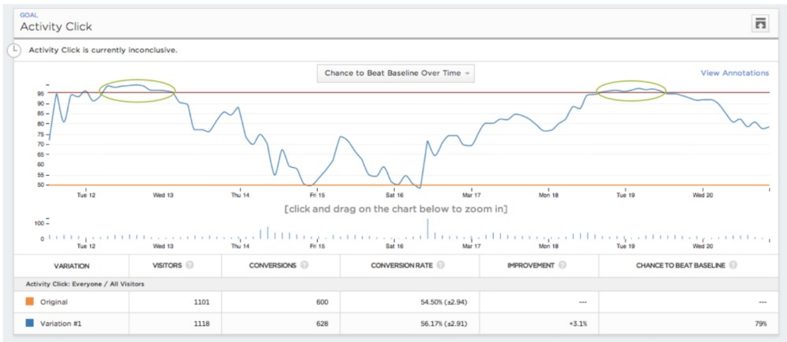

Figure 3. Continuous Monitoring of Static p-values

the sample, meaning that $p_n < \alpha$. By Example 3, that means that $X_n \in A_\alpha$ so that at time n we would reject the null hypothesis under the static hypothesis test at level α.

Theorem 4 (Law of Iterated Logarithm)**.** Suppose that $X_1, X_2, \ldots \overset{iid}{\sim} P$ are iid with $\mathbb{E}[X_1] = 0$ and $\mathrm{Var}(X_1) = \sigma^2$.

Then, we have that

$$\limsup_{n\to\infty} \frac{\sum_{i=1}^n X_i}{\sigma\sqrt{2n\log\log(n)}} = 1.$$

Proof.

See Theorem 8.5.2 in Durrett (2019). //

Corollary 5 (Continuously Monitoring Hypothesis is Bad)**.** Suppose that $X_1, X_2, ... \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$. And we wish to test $N : \theta = 0$ versus $A : \theta \neq 0$. Suppose we wish to conduct the test at significance level $\alpha = 0.05$. At any $n \in \mathbb{N}$, the uniformly most powerful test is

$$\delta_\alpha^{(n)}(\{X\}_{i=1}^n) = \begin{cases} N, & \text{if } \left|\frac{1}{\sigma\sqrt{n}}\sum_{i=1}^n X_i\right| \leq 1.96 \\ A, & \text{if } \left|\frac{1}{\sigma\sqrt{n}}\sum_{i=1}^n X_i\right| > 1.96. \end{cases}$$

Suppose that now we consider a test $\delta^{(\infty)} : \mathbb{R}^\infty \to \{N, A\}$ where

$$\delta^{(\infty)}(\{X_i\}_{i\in\mathbb{N}}) = \begin{cases} A, & \text{if } \exists n \in \mathbb{N} \text{ s.t. } \delta_n^{(\alpha)}(\{X_i\}_{i=1}^n) = A, \\ N, & \text{o.w.} \end{cases}$$

In other words, we favor the alternative hypothesis over the null hypothesis if there exists a static hypothesis test $\delta_\alpha^{(n)}$ at significance level α that concludes that $\delta_\alpha^{(n)}(\{X_i\}_{i=1}^n) = A$. We claim that $P_{\mathcal{N}(0,\sigma^2)}(\delta^{(\infty)} = A) = 1$. In other words, the probability of a Type I error is 1!

Proof.

By Theorem 4, we have that

$$\limsup_{n\to\infty} \frac{\sum_{i=1}^n X_i}{\sigma\sqrt{n}} = \infty$$

meaning that $\left|\frac{1}{\sigma\sqrt{n}}\sum_{i=1}^n X_i\right| > 1.96$ infinitely often almost surely. In other words, with probability 1, the test statistic will break the threshold 1.96 infinitely many times under the null hypothesis. //

It's a very natural question to ask, what kind of control do we have over the Type I error if we accumulate data points $X_1, X_2, ...$ in a hypothesis test and at each $n \in \mathbb{N}$, we run a static hypothesis test at level α? Corollary 5 presents a simple argument that shows that if we're willing to accumulate an infinite sample, our Type I error rate via continuously monitoring will be 1 so that in fact we have no control of it! Figure 2 taken from Johari et al. (2016) shows that in finite sample we will have Type I error rates that are larger than desired.

Given these negative results, a procedure where static p-values are continuously monitored is a bad way to conduct hypothesis tests. The question "Always Valid Inference" seeks to answer is, is there a way to control Type I error with test results being continuously monitored? We give some reasons for why this is an important question:

(a) It's very tempting for industry practitioners to continuously check static p-values.

(b) It's costly to wait extra time to conclude an A/B test if one hypothesis is looking much better than another and there are more things that want to be experimented with.

(c) It's good for an employee's career to find statistically significant results so they might be tempted to p-hack.

(d) Different employees at a company might be looking at a dashboard with the results of these tests and they may want to call the test either way at any time.

2.2 Some Definitions and Initial Results

We suppose that we have data $X = (X_n)_{n\in\mathbb{N}} \overset{iid}{\sim} P_\theta$ where $\theta \in \Theta \subset \mathbb{R}^P$ with Θ open. We let $(\mathcal{F}_n)_{n\in\mathbb{N}}$ be the filtration generated by $(X_n)_{n\in\mathbb{N}}$. We focus on testing

- $N : \theta = \theta_0$
- $A : \theta \neq \theta_0$

Definition 6 (Decision Rule)**.** A pair (T, δ) is a *decision rule* when T is a stopping time with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$ that denotes the sample size at which the test ends and $\delta : (X_n)_{n=1}^T \to \{N, A\}$ is a test that denotes whether the null hypothesis is favored to the alternative hypothesis or the alternative hypothesis is favored to the null hypothesis.

When T is not data-dependent, we have a static hypothesis testing problem like those covered in Section 1.

Definition 7 (Sequential Tests)**.** A *sequential test* is a family of decision rules $(T_\alpha, \delta_\alpha)_{\alpha\in(0,1)}$ such that

(1) The decision rules are nested in the sense that T_α is almost surely nonincreasing in α and δ_α is almost surely nondecreasing in α.

(2) For each $\alpha \in (0, 1)$, $P_{\theta_0}(\delta_\alpha = A) \leq \alpha$.

We make some comments:

(a) How should we understand the decision rule? T after every sample X_n, says do we stop now and decide which hypothesis to favor or do we wait at least another sample to make the decision. δ favors a hypothesis when forced to.

(b) Recall the purpose of $\alpha \in (0, 1)$, which controls the Type I error rate. A higher α means that we're more okay incorrectly favoring the alternative hypothesis to the null hypothesis. What does it mean for T_α to be almost surely nonincreasing in α? That means that if at a given level we have decided to make a decision, then at any more permissible level, we should also have already made a decision. What does it mean for δ_α to be nondecreasing in α? For a given sample and given level, if we favor the alternative hypothesis to the null hypothesis, then at any more permissible level we should also favor the alternative hypothesis to the null hypothesis.

(c) Why do we care about the condition that $P_{\theta_0}(\delta_\alpha = A) \leq \alpha$? This condition is saying that the Type I error of δ_α is less than or equal to α.

Definition 8 (Always Valid p-value Process)**.** A sequence of static p-values $(p_n)_{n\in\mathbb{N}}$ is an *always valid p-value process* if given any (possibly infinite) stopping time T with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}}$, it holds that:

$$\forall s \in [0, 1],\ P_{\theta_0}(p_T \leq s) \leq s$$

Theorem 9 (Equivalence Between Sequential Test and Always Valid p-value Process)**.** Let $(T_\alpha, \delta_\alpha)_{\alpha\in(0,1)}$ be a sequential test. Then (i)

$$p_n := \inf\{\alpha \in [0, 1] : T_\alpha \leq n, \delta_\alpha = 1\}$$

defines an always valid p-value process. In addition, given an always valid p-value process $(p_n)_{n\in\mathbb{N}}$, (ii) a sequential test $(\tilde{T}_\alpha, \tilde{\delta}_\alpha)_{\alpha\in(0,1)}$ is obtained from $(p_n)_{n\in\mathbb{N}}$ as follows:

$$\tilde{T}_\alpha := \inf\{n \in \mathbb{N} : p_n \leq \alpha\},\ \tilde{\delta}_\alpha := \mathbb{1}_{\{\tilde{T}_\alpha<\infty\}}.$$

If $(p_n)_{n\in\mathbb{N}}$ was derived as in (i) and $\delta_\alpha = 0 \implies T_\alpha = \infty$, then $(T_\alpha, \delta_\alpha)_{\alpha\in(0,1)} = (\tilde{T}_\alpha, \tilde{\delta}_\alpha)_{\alpha\in(0,1)}$.

Proof.

[i]:

Nestedness of $(T_\alpha, \delta_\alpha)_{\alpha\in(0,1)}$ implies that

$$\begin{aligned} &\{p_n \leq s\} \subseteq \{T_{s+\epsilon} \leq n, \delta_{s+\epsilon} = 1\} \subseteq \{\delta_{s+\epsilon} = 1\} \\ \implies &P_{\theta_0}(p_T \leq s) \leq P_{\theta_0}(\cup_{n\in\mathbb{N}}\{p_n \leq s\}) \leq P_{\theta_0}(\delta_{s+\epsilon} = 1) \leq s + \epsilon. \end{aligned}$$

Sending $\epsilon \downarrow 0$ and transitivity gives the result.

[ii]:

It's plain to see that the tests are nested by definition of $\tilde{T}_\alpha$ and $\tilde{\delta}_\alpha$ and that $\tilde{T}_\alpha = \infty \implies \tilde{\delta}_\alpha = 0$. For any $\epsilon > 0$, we have that

$$\begin{aligned} P_{\theta_0}(\tilde{\delta}_\alpha = 1) &= P_{\theta_0}(\tilde{T}_\alpha < \infty) \\ &\leq P_{\theta_0}(p_{\tilde{T}_\alpha} \leq \alpha + \epsilon) \\ &\leq \alpha + \epsilon. \end{aligned}$$

Sending $\epsilon \downarrow 0$ and transitivity give the result. □

2.3 The mSPRT

In this section, we introduce the *mixture sequential probability ratio tests* (mSPRT) proposed in Johari et al. (2022). This method provides a simple construction of an always valid p-value process designed to give an efficient trade-off between giving users the ability to detect true effects with maximum probability while minimizing the length of experiments.

We restrict $\{P_\theta\}_{\theta\in\Theta}$ from the start of Section 2.2 to be a single parameter exponential family (ie., $f_\theta(x) = F'_\theta(x) = f_0(x)\exp(\theta x - \psi(\theta))$). We further assume Θ is an open interval.

The mSPRT takes as input a mixing distribution H over Θ, which we restrict to have everywhere continuous and positive derivative. Given a sample average $S_n := n^{-1}\sum_{i=1}^n X_i$, we define the *mixture likelihood ratio* with respect to H to be

$$\begin{aligned} \Lambda_n^H(\{X_i\}_{i=1}^n) &:= \int_\Theta \prod_{i=1}^n \left(\frac{f_0(X_i)\exp(\theta X_i - \psi(\theta))}{f_0(X_i)\exp(\theta_0 X_i - \psi(\theta_0))}\right) dH(\theta) \\ &= \int_\Theta \left(\frac{f_0(S_n)}{f_0(S_n)}\right)^n \left(\frac{\exp\left(\theta\sum_{i=1}^n X_i - n\psi(\theta)\right)}{\exp\left(\theta_0\sum_{i=1}^n X_i - n\psi(\theta_0)\right)}\right) dH(\theta) \\ &= \int_\Theta \left(\frac{f_\theta(S_n)}{f_{\theta_0}(S_n)}\right)^n dH(\theta) \\ &=: \Lambda_n^H(S_n) \end{aligned}$$

The mSPRT is then defined by

$$\begin{aligned} T_\alpha^H &:= \inf\{n \in \mathbb{N} : \Lambda_n^H(S_n) \geq \alpha^{-1}\}, \\ \delta_\alpha^H &:= \begin{cases} A, & \text{if } T_\alpha^H < \infty \\ N, & \text{otherwise} \end{cases} \end{aligned}$$

We claim using standard Martingale techniques that the Type I error is controlled at level α.

Lemma 10 (mSPRT is Martingale)**.** Consider the definition of the mSPRT $\Lambda_n^H(S_n)$ given above with $\Lambda_0^H(S_0) := 1$. We have that $(\Lambda_n^H(S_n))_{n\in\mathbb{N}}$ is a martingale relative to the filtration $\mathcal{F}_n := \sigma(X_1, ..., X_n)$ under the measure of the null hypothesis $P_{\theta_0}^{\otimes n}$.

Proof.

We have that for any $n \in \mathbb{N}$,

- Trivially, $\Lambda_n^H(S_n) \in \mathcal{F}_n$.

- The expectation is finite at any $n \in \mathbb{N}$,

$$\begin{aligned}
\mathbb{E}_{P_{\theta_0}^{\otimes n}}\left[\left|\Lambda_n^H(S_n)\right|\right] &= \mathbb{E}_{P_{\theta_0}^{\otimes n}}\left[\int_\Theta \left(\frac{f_\theta(S_n)}{f_{\theta_0}(S_n)}\right)^n dH(\theta)\right] \\
&= \int_\Theta \mathbb{E}_{P_{\theta_0}^{\otimes n}}\left[\left(\frac{f_\theta(S_n)}{f_{\theta_0}(S_n)}\right)^n\right] dH(\theta), \text{ by Tonelli} \\
&= \int_\Theta \prod_{i=1}^n \underbrace{\mathbb{E}_{P_{\theta_0}}\left[\frac{f_\theta(X_i)}{f_{\theta_0}(X_i)}\right]}_{=1} dH(\theta), \text{ by } \perp\!\!\!\perp [X_1, ..., X_n] \\
&= 1 \\
&< \infty
\end{aligned}$$

- The martingale property is satisfied,

$$\begin{aligned}
\mathbb{E}_{P_{\theta_0}}\left[\Lambda_n^H(S_n) \mid \mathcal{F}_{n-1}\right] &= \mathbb{E}_{P_{\theta_0}}\left[\int_\Theta \left(\frac{f_\theta(S_n)}{f_{\theta_0}(S_n)}\right)^n dH(\theta) \mid \mathcal{F}_{n-1}\right] \\
&= \Lambda_{n-1}^H(S_{n-1})\mathbb{E}_{P_{\theta_0}}\left[\int_\Theta \frac{f_\theta(X_n)}{f_{\theta_0}(X_n)} dH(\theta)\right], \text{ after some algebra} \\
&= \Lambda_{n-1}^H(S_{n-1}) \int_\Theta \underbrace{\mathbb{E}_{P_{\theta_0}}\left[\frac{f_\theta(X_n)}{f_{\theta_0}(X_n)}\right]}_{=1} dH(\theta), \text{ by Tonelli} \\
&= \Lambda_{n-1}^H(S_{n-1}).
\end{aligned}$$

□

Lemma 11 (Maximal Inequality for Martingales)**.** Let $(M_n)_{n\in\mathbb{N}}$ be a non-negative martingale relative to $(\mathcal{F}_n)_{n\in\mathbb{N}}$. Then, we have that for any $\lambda > 0$

$$\Pr\left(\max_{i\in[0:n]} M_i \geq \lambda\right) \leq \frac{\mathbb{E}[M_0]}{\lambda}$$

Proof.

We define the stopping time

$$T := \begin{cases} \min\{0 \leq k \leq n : M_k \geq \lambda\}, & \text{if } M_k \geq \lambda \text{ for some } k \\ n, & \text{otherwise.} \end{cases}$$

Since $T \leq n$, by the optional stopping theorem, we have that

$$\begin{aligned}\mathbb{E}[M_0] &= \mathbb{E}[M_T] \\ &\geq \mathbb{E}\left[M_T \mathbb{1}_{\{\max_{0\leq k\leq n} M_k \geq \lambda\}}\right] \\ &\geq \lambda \Pr\left(\max_{0\leq k\leq n} M_k \geq \lambda\right)\end{aligned}$$

□

Theorem 12 (mSPRT has Type I Error Rate Controlled)**.** Consider the definition of the mSPRT $\Lambda_n^H(S_n)$ given above. We have that $P_{\theta_0}(\delta_\alpha^H = A) \leq \alpha$.

Proof.

By Lemma 10, we have that $\Lambda_n^H(S_n)$ with $\Lambda_0^H(S_0) := 1$ is a martingale. Applying Lemma 11, we have that

$$\begin{aligned}P_{\theta_0}\left(\max_{0\leq k\leq n} \Lambda_k^H(S_k) \geq \alpha^{-1}\right) &\leq \mathbb{E}_{P_{\theta_0}}[\Lambda_0^H(S_0)]\alpha \\ &= \alpha.\end{aligned}$$

We then have that

$$\begin{aligned}P_{\theta_0}\left(\delta_\alpha^H = A\right) &= P_{\theta_0}(T_\alpha^H < \infty) \\ &= P_{\theta_0}\left(\max_{n\in\mathbb{N}} \Lambda_n^H(S_n) \geq \alpha^{-1}\right) \\ &= \lim_{n\to\infty} P_{\theta_0}\left(\max_{0\leq k\leq n} \Lambda_k^H(S_k)\right), \\ &\qquad \text{by continuity of probability measures and that } \left\{\max_{0\leq k\leq n} \Lambda_k^H(S_k)\right\} \uparrow \left\{\max_{n\in\mathbb{N}} \Lambda_n^H(S_n)\right\} \\ &\leq \alpha.\end{aligned}$$

□

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